```
<!DOCTYPE html>
<html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1,viewport-fit=cover">
<title>Blocky Character Rig</title>
<style>
:root{--bg1:#5aa9ff;--bg2:#e4f3ff;--panel:rgba(255,255,255,.93);--ink:#18212f;--line:#c9d4e6;--acc:#2f6bff}
@media (prefers-color-scheme:dark){:root{--bg1:#16203a;--bg2:#3b4c7e;--panel:rgba(22,28,44,.94);--ink:#eef2fb;--line:#3a4764;--acc:#86a8ff}}
*{box-sizing:border-box}
html,body{margin:0;height:100%;overflow:hidden;font-family:system-ui,-apple-system,"Segoe UI",sans-serif;background:linear-gradient(var(--bg1),var(--bg2));color:var(--ink)}
canvas{display:block;width:100%;height:100%;touch-action:none}
#ui{position:fixed;left:8px;right:8px;bottom:calc(8px + env(safe-area-inset-bottom,0px));max-width:680px;margin:0 auto;padding:10px;border-radius:14px;background:var(--panel);display:grid;grid-template-columns:1fr 1fr;gap:8px 10px;box-shadow:0 6px 24px rgba(0,0,0,.25)}
label{font-size:12px;font-weight:600;display:block;margin-bottom:3px}
label span{opacity:.6;font-weight:400}
.r{display:flex;gap:4px}
select,button{font:inherit;font-size:14px;color:var(--ink);background:transparent;border:1px solid var(--line);border-radius:8px;height:34px}
select{flex:1;min-width:0;padding:0 4px;background:var(--panel)}
.r button{width:32px;padding:0;font-size:18px}
#bar{grid-column:1/3;display:flex;gap:8px}
#bar button{flex:1;font-weight:600}
#bar button.on{background:var(--acc);color:#fff;border-color:var(--acc)}
button:focus-visible,select:focus-visible{outline:2px solid var(--acc);outline-offset:1px}
#hint{position:fixed;top:calc(10px + env(safe-area-inset-top,0px));left:0;right:0;text-align:center;font-size:12px;opacity:.7;pointer-events:none}
#faces{grid-column:1/3;display:flex;gap:6px;overflow-x:auto;padding:2px 2px 6px;scrollbar-width:thin}
#faces button{flex:none;width:48px;height:48px;padding:2px;border-radius:12px;background:var(--sk,#f2c29b)}
#faces canvas{display:block;width:100%;height:100%}
#faces button.on{outline:3px solid var(--acc);outline-offset:1px}
</style></head><body>
<div id="hint">Tap a face below · drag to orbit · pinch to zoom</div>
<div id="ui"></div>
<script src="https://cdnjs.cloudflare.com/ajax/libs/three.js/r128/three.min.js"></script>
<script>
const clamp=(x,a,b)=>Math.max(a,Math.min(b,x)),$=s=>document.querySelector(s),sn=Math.sin;
function RB(w,h,d,r=.16,s=12){const g=new THREE.BoxGeometry(w,h,d,s,s,s),p=g.attributes.position,n=g.attributes.normal,hw=w/2-r,hh=h/2-r,hd=d/2-r;
const rm=(v,half,inner)=>{const a=Math.abs(v)/half,sg=Math.sign(v);return sg*(a<=.5?a/.5*inner:inner+(a-.5)/.5*r)};
for(let i=0;i<p.count;i++){const x=rm(p.getX(i),w/2,hw),y=rm(p.getY(i),h/2,hh),z=rm(p.getZ(i),d/2,hd);
const cx=clamp(x,-hw,hw),cy=clamp(y,-hh,hh),cz=clamp(z,-hd,hd);let dx=x-cx,dy=y-cy,dz=z-cz;const L=Math.hypot(dx,dy,dz)||1;dx/=L;dy/=L;dz/=L;
p.setXYZ(i,cx+dx*r,cy+dy*r,cz+dz*r);n.setXYZ(i,dx,dy,dz)}return g}
const gc={},G=(w,h,d,r)=>{const k=[w,h,d,r]+'';return gc[k]||(gc[k]=RB(w,h,d,r))};
function limbGeo(w,h,d,r){const g=RB(w,h,d,r,24),I=g.index.array,p=g.attributes.position.array,T=[],B=[];
for(let i=0;i<I.length;i+=3){((p[I[i]*3+1]+p[I[i+1]*3+1]+p[I[i+2]*3+1])>=0?T:B).push(I[i],I[i+1],I[i+2])}
g.setIndex(T.concat(B));g.clearGroups();g.addGroup(0,T.length,0);g.addGroup(T.length,B.length,1);
g.userData={own:1,a:0,p0:p.slice(),n0:g.attributes.normal.array.slice()};return g}
function bend(g,a){const u=g.userData;if(Math.abs(a-u.a)<.003)return;u.a=a;const P=g.attributes.position,N=g.attributes.normal;
for(let i=0;i<P.count;i++){const j=i*3,y=u.p0[j+1],e=Math.min(1,Math.max(0,(y+.3)/.5)),t=1-e*e*(3-2*e),c=Math.cos(a*t),n=Math.sin(a*t);
P.array[j]=u.p0[j];P.array[j+1]=y*c-u.p0[j+2]*n;P.array[j+2]=y*n+u.p0[j+2]*c;
N.array[j]=u.n0[j];N.array[j+1]=u.n0[j+1]*c-u.n0[j+2]*n;N.array[j+2]=u.n0[j+1]*n+u.n0[j+2]*c}
P.needsUpdate=N.needsUpdate=true}
const P=c=>new THREE.MeshPhysicalMaterial({color:c,roughness:.4,clearcoat:.5,clearcoatRoughness:.25});
const M=(g,m,x=0,y=0,z=0)=>{const o=new THREE.Mesh(g,m);o.position.set(x,y,z);o.castShadow=o.receiveShadow=true;return o};

/* ---------- DATA ---------- */
// name, skin, hair color, hair style(0-5), width, height, head size
const CH=[["Bob","#f2c29b","#3b2a1c",1,1,1,1],["Luna","#e8b48a","#1a1a24",2,1,1,1],["Max","#8d5a3b","#111111",1,1,1,1],["Zoe","#ffd9b8","#e0a030",3,1,1,1.03],["Kai","#c68642","#222222",5,1,1,1],["Mia","#f5cba7","#b5451b",4,1,1,1],["Rex (zombie)","#6b8f3a","#2a3a1a",0,1,1,1],["Blu (alien)","#4fa8ff","#ffffff",0,1,1,1.12],["Pinky","#ff9ec4","#ff4f9a",2,1,1,1],["Goldie","#f6c945","#a06a00",1,1,1,1],["Shadow","#3a3a46","#111111",5,1,1,1],["Frost","#cfeaff","#7ad0ff",3,1,1,1],["Mint","#8fe3b0","#2f8f5a",4,1,1,1],["Lava","#e8573a","#4a0f00",5,1,1,1],["Cocoa","#5a3825","#000000",2,1,1,1],["Violet","#b58cff","#4a2a8a",3,1,1,1],["Ghost","#f0f0f5","#c9c9d9",0,1,1,1.05],["Tank","#d9a077","#6b3a1a",1,1.22,1,1],["Tiny","#f7d1b0","#8a5a2b",4,.7,.7,1.2],["Giant","#b98560","#222222",5,1.05,1.3,.92],["Grandma","#f5cba7","#dcdce6",7,1,.93,1,30],["Grampa","#efc29b","#e9e9f0",6,1.06,.95,1,31],["Officer Jack","#f2c29b","#2a1a10",1,1,1,1,32],["Officer Jill","#e8b48a","#5a3418",2,1,1,1,33]];
// name, shirt, pants, shoes, pattern, patColor, hat, hatColor, longSleeve, shorts, extra, extraColor
const OU=[
["Plain Blue","#3b82f6","#2b2f42","#26262b","","","","",0,0],
["Red Logo Tee","#f5f5f5","#33384d","#26262b","logo","#e3302b","","",0,0],
["Sailor Stripes","#ffffff","#1e3a8a","#26262b","stripes","#1e3a8a","cap","#1e3a8a",0,0],
["Gray Hoodie","#7b8190","#3a3f4f","#26262b","pocket","#5a5f70","beanie","#3a3f4f",1,0],
["Lightning Hero","#ffd23f","#c1121f","#26262b","bolt","#c1121f","","",1,0,"cape","#c1121f"],
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["Star Pop","#ff6fb5","#6a3ea1","#26262b","star","#ffffff","","",0,0],
["Heart Tee","#ffffff","#ff5d8f","#26262b","heart","#ff3b6b","","",0,0],
["Business Suit","#2a2f3a","#2a2f3a","#111111","tie","#c1121f","","",1,0],
["Chef","#ffffff","#222222","#111111","buttons","#bbbbbb","chef","#ffffff",1,0],
["Wizard","#5b2a9e","#3a1a6e","#26262b","star","#ffd23f","wizard","#5b2a9e",1,0,"cape","#7a3fd1"],
["Knight","#aab4c4","#6b7587","#444444","band","#e3302b","helmet","#aab4c4",1,0,"cape","#c1121f"],
["Cowboy","#b5651d","#1e3a8a","#3b2412","buttons","#f5e1a4","cowboy","#7a4a1a",1,0],
["Royal King","#c1121f","#ffd23f","#26262b","band","#ffd23f","crown","#ffd23f",1,0,"cape","#8b0f1a"],
["Summer Smile","#ff9f43","#28c4c4","#26262b","smile","#ffe14d","","",0,1],
["Ninja","#1a1a22","#1a1a22","#111111","","","band","#e3302b",1,0,"scarf","#e3302b"],
["Pirate","#f1e9d2","#2a2a2a","#111111","stripes","#c1121f","top","#111111",1,0],
["Astronaut","#f2f4f8","#f2f4f8","#9aa4b5","logo","#1e5bd6","helmet","#f2f4f8",1,0,"pack","#9aa4b5"],
["Soccer Pro","#1fa05a","#ffffff","#111111","band","#ffffff","","",0,1],
["Basketball","#6a1b9a","#6a1b9a","#ffd23f","band","#ffd23f","","",0,1],
["Punk Rocker","#1a1a1a","#5b2a9e","#111111","bolt","#ff3bcf","","",0,0],
["Party Time","#ff4fa0","#3bd1ff","#26262b","star","#ffd23f","party","#ffd23f",0,0],
["Bunny Pajama","#ffd1e8","#ffd1e8","#ffffff","heart","#ff7ab8","ears","#ffd1e8",1,0],
["Gardener","#7bd34a","#8a5a2b","#3b2412","pocket","#5aa02a","cap","#d8b36b",0,0],
["Lifeguard","#ff3b30","#ff3b30","#26262b","band","#ffffff","cap","#ffffff",0,1],
["Scientist","#ffffff","#3a4a5a","#222222","pocket","#9aa4b5","","",1,0,"glasses"],
["Tourist","#29b6a8","#f1d9a0","#26262b","heart","#ffcf3f","cap","#ffffff",0,1,"glasses"],
["Candy Stripe","#ffffff","#ff4f6e","#26262b","stripes","#ff4f6e","","",0,0],
["Frosty","#bfe6ff","#4a90d9","#26262b","buttons","#ffffff","beanie","#ffffff",1,0,"scarf","#e3302b"],
["Golden Hero","#ffd23f","#ffb300","#26262b","star","#ffffff","crown","#ff9f1c",0,0,"cape","#ff9f1c"],
["Shadow Hoodie","#15151c","#15151c","#111111","pocket","#2a2a36","beanie","#15151c",1,0,"glasses"],["Grandma
   Cardigan","#d98cb3","#6b5a7a","#5a4636","buttons","#f5e1a4","","",1,0,"glasses"],["Grampa
   Sweater","#5f86a8","#8a7a5a","#3b2412","stripes","#e8e0c8","","",1,0,"glasses"],
["Police Officer","#2f6fd6","#16295a","#111111","badge","#ffd23f","police","#1b3a7a",1,0],
["Police Woman","#3b7be0","#1b2f66","#111111","badge","#ffd23f","police","#1b3a7a",1,0]];
/* eyes: o oval, d dot, r ring, p ring+pupil, w wide ring, g > < wink, ^ happy, u closed, - line, h half-lid,
   H heart, S star, @ spiral, x cross
brows: n none, f flat, a angry, s sad, u raised, za/zs zigzag (angry/sad)
mouths: c curve(p), sm smirk, cat, zig, grin(p=teeth), tongue, box, wavy, o(p=size), sob, teeth
extras: k gaze, ec eye color, bl blush, tr tears, jt joy tears, sw sweat, zz sleep, sl shock lines, vn anger
   vein, sh shades */
const EM=[["Neutral",["d","d"],["n","n"],["c",.25]],["Happy",["o","o"],["u","u"],["grin",0]],["Talking",["o","
   o"],["n","n"],["talk",.3]],["Scared",["r","r"],["n","n"],["wavy"]],["Worried",["p","p"],["s","s"],["zig"],{k:[
   0,.5]}],["Thinking",["p","p"],["f","u"],["c",-.1],{k:[.8,-
   .8]}],["Skeptical",["h","o"],["a","u"],["sm"]],["Serious",["d","d"],["f","f"],["c",0]],["Angry",["o","o"],["a"
   ,"a"],["teeth"],{ec:"#e8261c",vn:1}],["Surprised",["w","w"],["u","u"],["o",1]],
["Sad",["p","p"],["s","s"],["c",-
   1],{k:[0,.6]}],["Crying",["p","p"],["s","s"],["sob"],{tr:1,k:[0,.6]}],["Laughing",["^","^"],["u","u"],["grin",
   1],{jt:1}],["Excited",["S","S"],["u","u"],["grin",1],{bl:1}],["Smirk",["d","h"],["f","u"],["sm"]],["Wink",["g"
   ,"o"],["n","n"],["c",1]],["Wink Tongue",["g","o"],["n","n"],["tongue"]],["Sleepy",["h","h"],["f","f"],["o",.2]
   ],["Sleeping",["u","u"],["n","n"],["c",.3],{zz:1}],["Love",["H","H"],["n","n"],["grin",0],{bl:1}],
["Starstruck",["S","S"],["u","u"],["grin",1]],["Embarrassed",["d","d"],["s","s"],["zig"],{bl:1,sw:1,k:[-
   .8,.5]}],["Disgusted",["g","g"],["a","a"],["zig"]],["Confused",["p","p"],["u","s"],["zig"],{k:[.7,-
   .4]}],["Bored",["h","h"],["f","f"],["c",0]],["Smug",["h","h"],["u","f"],["sm"],{bl:1}],["Evil",["h","h"],["a",
   "a"],["grin",1],{ec:"#c4141c"}],["Shocked",["w","w"],["u","u"],["box"]],["Shock
   Lines",["p","p"],["za","za"],["box"],{sl:1}],["Terrified",["w","w"],["zs","zs"],["wavy"],{sw:1}],
["Joyful",["^","^"],["u","u"],["grin",0],{bl:1}],["Silly",["w","d"],["u","n"],["tongue"]],["Tongue Out",["d","
   d"],["n","n"],["tongue"],{bl:1}],["Proud",["^","^"],["u","u"],["c",.8]],["Determined",["o","o"],["a","a"],["te
   eth"]],["Nervous",["p","p"],["zs","zs"],["zig"],{sw:1,k:[.5,.2]}],["Dizzy",["@","@"],["n","n"],["wavy"]],["Ann
   oyed","h","h"],["a","a"],["c",0]],["Jealous",["h","h"],["a","f"],["c",-
   .6]],["Hopeful",["p","p"],["s","s"],["c",.6],{k:[0,-.7]}],["Pouting",["p","p"],["s","s"],["o",.1],{bl:1}],
["Grumpy",["h","h"],["a","a"],["c",-
   1]],["Mischievous",["h","h"],["a","u"],["cat"]],["Relieved",["u","u"],["n","n"],["c",1],{bl:1}],["Cool",[0,0],
   [0,0],["sm"],{sh:1}],["Uncanny
   Grin",["e","e"],["m","m"],["big",1]],["Glam",["m","m"],["m","m"],["lips"],{ec:"#6b3a22"}],["Man Face",["l","l"
   ],["mr","m"],["big",0]],["Dead",["x","x"],["n","n"],["tongue"]],["Devil",["h","h"],["a","a"],["grin",2],{ec:"#
   c4141c",hn:1}],["Angel",["d","d"],["u","u"],["c",.8],{ha:1,bl:1}],["Pirate",["d","p"],["a","u"],["grin",1],{ep
   :1}],["Fancy",["d","p"],["u","u"],["c",.3],{ms:1,mn:1}],["Geek",["p","p"],["u","u"],["buck"],{gs:1}],["Scream"
   ,["g","g"],["a","a"],["o",1.5]],["Monster",["r","w"],["a","a"],["grin",2]],["Rockstar",[0,0],[0,0],["grin",1],
   {sh:"#c1121f"}],["Goofy",["p","d"],["u","n"],["buck"],{k:[.6,.2]}],["Grimace",["p","p"],["zs","zs"],["teeth"],
   {sw:1}],["Yawn",["u","u"],["f","f"],["o",1]]];
const
   PO=[["Idle",t=>[.08*sn(t*2),.1,-.08*sn(t*2),-.1,0,0,0,.06*sn(t*1.3),0,.025*sn(t*3.2),0,0,-.18,-.18,.03,.03]],
["Wave",t=>[0,.12,0,-2.55+.38*sn(t*11),0,0,0,.12,.1,.025*sn(t*3.2),0,0,-.2,-.5+.35*sn(t*11),.03,.03]],
["Walk",t=>{const w=sn(t*9),c=Math.cos(t*9);return[-.5*w,.1,.5*w,-.1,.6*w,-.6*w,0,0,0,Math.abs(w)*.1,0,0,-
   (.3+.4*Math.max(0,w)),-(.3+.4*Math.max(0,-w)),.08+.9*Math.max(0,-c),.08+.9*Math.max(0,c)]}],
["Run",t=>{const w=sn(t*14),c=Math.cos(t*14);return[-1.1*w,.15,1.1*w,-.15,w,-w,0,0,0,Math.abs(w)*.25,.2,0,-
   (1.2+.4*Math.max(0,w)),-(1.2+.4*Math.max(0,-w)),.2+1.5*Math.max(0,-c),.2+1.5*Math.max(0,c)]}],
["Cheer",t=>{const j=Math.abs(sn(t*6.5));return[0,2.7+.2*sn(t*13),0,-2.7-.2*sn(t*13),-.25*j,.25*j,-
   .15,0,.06*sn(t*13),j*.55,0,0,-.3+.2*sn(t*13),-.3-.2*sn(t*13),.1+.7*j,.1+.7*j]}],
["Think",t=>[-.8,-.55,-1,.45,0,0,-.16,-.22,.12+.03*sn(t*2),.02,0,0,-1.1,-2.1,.02,.02]],
["Shrug",t=>{const k=.5+.5*sn(t*3.4);return[-.6,.7+.1*k,-.6,-.7-.1*k,0,0,0,0,.22,.07*k,0,0,-.9,-.9,0,0]}],
["Point",t=>[-.15,1.45+.05*sn(t*3),-.25-.15*sn(t*2.5),-.15,0,0,0,.4,0,.02,0,0,-.1,-.25,.02,.02]],
["Scared",t=>[-1,.95,-1,-.95,.1,-.1,-.12,0,.05*sn(t*41),0,-.1,0,-1.4,-1.4,.5,.5]],
["Dance",t=>{const a=sn(t*6);return[-.3,1.6+.8*a,-.3,-1.6+.8*a,.4*a,-
   .4*a,0,0,.2*a,Math.abs(a)*.2,0,.5*sn(t*3),-.6-.5*a,-.6+.5*a,.3+.35*Math.abs(a),.3+.35*Math.abs(a)]}],
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["Pick Up",t=>{const u=.5+.5*sn(t*1.6);return[-.6*u,.1,-1.1*u,-.1,-.9*u,-.9*u,.3*u,0,0,-.7*u,.35*u,0,-.4*u,-.5*u,1.8*u,1.8*u]}],
["Cough",t=>{const k=Math.pow(Math.max(0,sn(t*3.2)),4);return[-.15,.1,-1,.45,0,0,.25*k,0,0,-.06*k,.25*k,0,-.2,-2.1,.05,.05]}]];

/* ---------- 2D FACE ---------- */
function heart(x,cx,cy,s){x.beginPath();x.moveTo(cx,cy+s*.9);x.bezierCurveTo(cx-s*1.6,cy-s*.2,cx-s*.8,cy-s*1.3,cx,cy-s*.4);x.bezierCurveTo(cx+s*.8,cy-s*1.3,cx+s*1.6,cy-s*.2,cx,cy+s*.9);x.closePath()}
function star(x,cx,cy,R,r){x.beginPath();for(let k=0;k<10;k++){const a=-Math.PI/2+k*Math.PI/5,q=k%2?r:R;x.lineTo(cx+Math.cos(a)*q,cy+Math.sin(a)*q)}x.closePath()}
const INK="#17110d",PK="#ff7a9a";
function ring(x,ex,ey,r,pu,k){x.fillStyle=INK;x.beginPath();x.arc(ex,ey,r+5,0,7);x.fill();const g=x.createRadialGradient(ex-r*.3,ey-r*.4,2,ex,ey,r);g.addColorStop(0,"#fff");g.addColorStop(1,"#c9ccd6");x.fillStyle=g;x.beginPath();x.arc(ex,ey,r,0,7);x.fill();
if(pu){const px=ex+k[0]*(r-pu-3),py=ey+k[1]*(r-pu-3);x.fillStyle=INK;x.beginPath();x.arc(px,py,pu,0,7);x.fill();x.fillStyle="#fff";x.beginPath();x.arc(px-pu*.3,py-pu*.3,pu*.28,0,7);x.fill()}}
function eye(x,t,ex,ey,sg,k,ec){x.strokeStyle=INK;x.fillStyle=ec;x.lineWidth=9;x.beginPath();
switch(t){
case"o":x.ellipse(ex,ey,10,24,0,0,7);x.fill();break;
case"d":x.ellipse(ex,ey,9,13,0,0,7);x.fill();break;
case"r":ring(x,ex,ey,23,0,k);break;
case"p":ring(x,ex,ey,23,10,k);break;
case"w":ring(x,ex,ey,28,5,k);break;
case"g":x.lineWidth=11;x.moveTo(ex-20*sg,ey-19);x.lineTo(ex+17*sg,ey);x.lineTo(ex-20*sg,ey+19);x.stroke();break;
case"^":x.moveTo(ex-18,ey+8);x.quadraticCurveTo(ex,ey-22,ex+18,ey+8);x.stroke();break;
case"u":x.moveTo(ex-18,ey-4);x.quadraticCurveTo(ex,ey+20,ex+18,ey-4);x.stroke();break;
case"-":x.moveTo(ex-17,ey+2);x.lineTo(ex+17,ey+2);x.stroke();break;
case"h":x.save();x.rect(ex-30,ey-1,60,40);x.clip();x.beginPath();x.ellipse(ex,ey,13,21,0,0,7);x.fill();x.restore();x.lineWidth=7;x.beginPath();x.moveTo(ex-16,ey-1);x.lineTo(ex+16,ey-1);x.stroke();break;
case"H":x.fillStyle="#ff3b6b";x.lineWidth=4;heart(x,ex,ey,14);x.fill();x.stroke();break;
case"S":x.fillStyle="#ffc400";x.lineWidth=4;star(x,ex,ey,27,12);x.fill();x.stroke();break;
case"@":x.lineWidth=4;for(let a=0;a<17;a+=.3)x.lineTo(ex+Math.cos(a)*a*1.5,ey+Math.sin(a)*a*1.5);x.stroke();break;
case"e":case"l":case"m":almond(x,t,ex,ey,sg,k,ec);break;
case"x":x.moveTo(ex-14,ey-14);x.lineTo(ex+14,ey+14);x.moveTo(ex+14,ey-14);x.lineTo(ex-14,ey+14);x.stroke()}}
function almond(x,t,ex,ey,sg,k,ec){const u=t==="e"?-26:t==="l"?-6:-18,d=t==="e"?20:t==="l"?14:16,o=ex-sg*30,i=ex+sg*30,path=()=>{x.beginPath();x.moveTo(o,ey+2);x.quadraticCurveTo(ex,ey+u,i,ey+2);x.quadraticCurveTo(ex,ey+d,o,ey+2);x.closePath()};
if(t==="m"){x.fillStyle="rgba(235,140,155,.5)";x.beginPath();x.ellipse(ex,ey-6,34,18,0,0,7);x.fill()}
path();x.fillStyle="#fff";x.fill();x.save();x.clip();const px=ex+k[0]*8,py=ey+2+k[1]*4;
x.fillStyle=t==="m"?ec:INK;x.beginPath();x.arc(px,py,t==="e"?14:t==="l"?11:13,0,7);x.fill();
if(t==="m"){x.fillStyle=INK;x.beginPath();x.arc(px,py,6,0,7);x.fill()}
x.fillStyle="#fff";x.beginPath();x.arc(px-4,py-4,3.5,0,7);x.fill();x.restore();
path();x.strokeStyle=INK;x.lineWidth=t==="m"?7:5;x.stroke();
if(t==="m"){x.beginPath();x.moveTo(o,ey+2);x.lineTo(o-sg*14,ey-6);x.stroke()}
else{x.lineWidth=3;x.beginPath();x.moveTo(o,ey-8);x.quadraticCurveTo(ex,ey+u-14,i,ey-8);x.stroke()}}
function brow(x,t,ex,sg){if(t==="n")return;const by=64,o=ex-sg*24,i=ex+sg*24;x.strokeStyle=INK;x.lineWidth=t[0]==="z"?4:9;x.beginPath();
if(t==="f"){x.moveTo(o,by);x.lineTo(i,by)}
else if(t==="a"){x.moveTo(o,by-8);x.lineTo(i,by+9)}
else if(t==="s"){x.moveTo(o,by+8);x.lineTo(i,by-9)}
else if(t==="u"){x.moveTo(o,by+4);x.quadraticCurveTo(ex,by-16,i,by+4)}
else if(t==="m"||t==="mr"){const h=t==="mr"?-14:0;x.lineWidth=12;x.moveTo(o,by+8+h);x.quadraticCurveTo(ex+sg*4,by-16+h,i,by+2+h)}
else{const d=t==="za"?1:-1;x.moveTo(o,by-10*d);x.lineTo(ex-sg*4,by);x.lineTo(ex+sg,by-4*d);x.lineTo(i,by+9*d)}
x.stroke()}
function FS(x,pth,fill,inner){pth();x.fillStyle=fill;x.fill();x.save();x.clip();inner&&inner();x.restore();pth();x.stroke()}
function mouth(x,m){const[t,p=0]=m,mx=128,my=172;x.lineWidth=7;x.strokeStyle=INK;x.beginPath();
if(t==="c"){x.moveTo(mx-30,my-p*8);x.quadraticCurveTo(mx,my+p*30,mx+30,my-p*8);x.stroke()}
else if(t==="sm"){x.moveTo(mx-26,my+4);x.quadraticCurveTo(mx+6,my+16,mx+34,my-10);x.stroke()}
else if(t==="cat"){x.moveTo(mx-38,my-2);x.quadraticCurveTo(mx-19,my+24,mx,my-2);x.quadraticCurveTo(mx+19,my+24,mx+38,my-2);x.stroke()}
else if(t==="zig"){x.moveTo(mx-36,my+4);for(let i=1;i<=6;i++)x.lineTo(mx-36+i*12,my+(i%2?-6:6)+4);x.stroke()}
else if(t==="grin")FS(x,()=>{x.beginPath();x.moveTo(mx-42,my-12);x.quadraticCurveTo(mx,my-2,mx+42,my-12);x.quadraticCurveTo(mx+34,my+42,mx,my+42);x.quadraticCurveTo(mx-34,my+42,mx-42,my-12);x.closePath()},"#4a1313",()=>{if(p===1){x.fillStyle="#fff";x.fillRect(mx-44,my-16,88,17)}if(p===2){x.fillStyle="#fff";[-30,30].forEach(dx=>{x.beginPath();x.moveTo(mx+dx-8,my-16);x.lineTo(mx+dx+8,my-16);x.lineTo(mx+dx,my+10);x.closePath();x.fill()})}x.fillStyle=PK;x.beginPath();x.ellipse(mx,my+44,24,15,0,0,7);x.fill()});
else if(t==="tongue"){x.strokeStyle="#c93a63";x.lineWidth=4;FS(x,()=>{x.beginPath();x.moveTo(mx-40,my-6);x.lineTo(mx+34,my+6);x.quadraticCurveTo(mx+10,my+30,mx-20,my+28);x.quadraticCurveTo(mx-44,my+20,mx-40,my-6);x.closePath()},PK,()=>{x.fillStyle="rgba(255,255,255,.35)";x.beginPath();x.ellipse(mx-12,my+4,16,5,.15,0,7);x.fill()})}
else if(t==="box")FS(x,()=>{x.beginPath();x.moveTo(mx-28,my-12);x.lineTo(mx+28,my-
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12);x.quadraticCurveTo(mx+42,my-12,mx+42,my+2);x.lineTo(mx+46,my+22);x.quadraticCurveTo(mx+48,my+36,mx+34,my+36);x.lineTo(mx-34,my+36);x.quadraticCurveTo(mx-48,my+36,mx-46,my+22);x.lineTo(mx-42,my+2);x.quadraticCurveTo(mx-42,my-12,mx-28,my-12);x.closePath()},"#c94e5c",()=>{x.fillStyle="#e87d89";x.fillRect(mx-50,my-16,100,26)});
else if(t==="wavy")FS(x,()=>{x.beginPath();x.moveTo(mx-50,my+4);x.bezierCurveTo(mx-30,my-14,mx-10,my+10,mx+12,my-2);x.bezierCurveTo(mx+30,my-12,mx+44,my-8,mx+50,my-6);x.bezierCurveTo(mx+52,my+24,mx+20,my+34,mx,my+30);x.bezierCurveTo(mx-30,my+34,mx-54,my+26,mx-50,my+4);x.closePath()},"#8c141a",()=>{x.fillStyle="rgba(255,255,255,.12)";x.fillRect(mx-60,my-16,120,16)});
else if(t==="o"||t==="talk"){const rx=t==="talk"?20:10+p*14,ry=t==="talk"?5+p*55:12+p*20;FS(x,()=>{x.beginPath();x.ellipse(mx,my+4+ry*.6,rx,ry,0,0,7)},"#4a1313",()=>{x.fillStyle=PK;x.beginPath();x.ellipse(mx,my+4+ry*1.4,rx*.8,ry*.5,0,0,7);x.fill()})}
else if(t==="big")FS(x,()=>{x.beginPath();x.moveTo(mx-64,my-24);x.quadraticCurveTo(mx,my-4,mx+64,my-24);x.quadraticCurveTo(mx+52,my+52,mx,my+54);x.quadraticCurveTo(mx-52,my+52,mx-64,my-24);x.closePath()},"#fff",()=>{x.lineWidth=4;x.beginPath();x.moveTo(mx-64,my+6);x.quadraticCurveTo(mx,my+22,mx+64,my+6);[-2,-1,0,1,2].forEach(i=>{x.moveTo(mx+i*20,my-10);x.lineTo(mx+i*20,my+10)});x.stroke()});
else if(t==="lips"){x.strokeStyle="#7a0a22";x.lineWidth=4;FS(x,()=>{x.beginPath();x.moveTo(mx-50,my+4);x.quadraticCurveTo(mx-24,my-22,mx,my-10);x.quadraticCurveTo(mx+30,my-26,mx+52,my-8);x.quadraticCurveTo(mx+34,my+34,mx-4,my+34);x.quadraticCurveTo(mx-40,my+30,mx-50,my+4);x.closePath()},"#d11a40",()=>{x.fillStyle="#fff";x.beginPath();x.moveTo(mx-38,my+6);x.quadraticCurveTo(mx,my-6,mx+40,my-2);x.quadraticCurveTo(mx+28,my+22,mx-4,my+22);x.quadraticCurveTo(mx-30,my+20,mx-38,my+6);x.fill();x.fillStyle="rgba(255,255,255,.55)";x.beginPath();x.ellipse(mx-10,my+28,9,3,0,0,7);x.fill()})}
else if(t==="buck"){x.moveTo(mx-30,my-6);x.quadraticCurveTo(mx,my+26,mx+30,my-6);x.stroke();x.fillStyle="#fff";x.beginPath();x.rect(mx-14,my+9,28,20);x.fill();x.lineWidth=5;x.stroke();x.beginPath();x.moveTo(mx,my+9);x.lineTo(mx,my+29);x.stroke()}
else if(t==="sob")FS(x,()=>{x.beginPath();x.moveTo(mx-36,my+22);x.quadraticCurveTo(mx,my-24,mx+36,my+22);x.quadraticCurveTo(mx,my+8,mx-36,my+22);x.closePath()},"#4a1313");
else if(t==="teeth"){x.fillStyle="#fff";x.rect(mx-36,my-4,72,28);x.fill();x.stroke();x.lineWidth=4;x.beginPath();x.moveTo(mx-12,my-4);x.lineTo(mx-12,my+24);x.moveTo(mx+12,my-4);x.lineTo(mx+12,my+24);x.moveTo(mx-36,my+10);x.lineTo(mx+36,my+10);x.stroke()}}
function drawFace(x,E,blink,mo,gl){const[,ey,bw,mt,X={}]=E,k=X.k||[0,0];x.clearRect(0,0,256,256);x.lineCap=x.lineJoin="round";
if(!X.sh)[[88,1],[168,-1]].forEach(([ex,sg],i)=>{let t=ey[i];if(blink&&"odrpwhelm".includes(t))t="-";eye(x,t,ex,112,sg,k,X.ec||INK);brow(x,bw[i],ex,sg)});
mouth(x,mo||mt);
if(X.bl){x.fillStyle="rgba(255,90,110,.5)";[48,208].forEach(bx=>{x.beginPath();x.ellipse(bx,148,18,10,0,0,7);x.fill()})}
if(X.tr){x.fillStyle="#5bc8ff";[88,168].forEach(bx=>{x.beginPath();x.ellipse(bx,146,7,13,0,0,7);x.fill()})}
if(X.jt){x.fillStyle="#5bc8ff";[[34,96,.6],[222,96,.6]].forEach(([bx,by,r])=>{x.beginPath();x.ellipse(bx,by,6,12,r,0,7);x.fill()})}
if(X.sw){x.fillStyle="#7fd4ff";x.beginPath();x.moveTo(214,40);x.quadraticCurveTo(230,64,214,74);x.quadraticCurveTo(198,64,214,40);x.fill()}
if(X.zz){x.fillStyle=INK;x.font="bold 34px sans-serif";x.fillText("Z",190,60);x.font="bold 22px sans-serif";x.fillText("z",224,34)}
if(X.sl){x.strokeStyle="#4a78c9";x.lineWidth=6;[[40,64],[54,80],[68,56],[82,72]].forEach(([lx,ly])=>{x.beginPath();x.moveTo(lx,10);x.lineTo(lx,ly);x.stroke()})}
if(X.vn){x.strokeStyle="#e8261c";x.lineWidth=6;[[-1,-1],[1,-1],[-1,1],[1,1]].forEach(([a,b])=>{x.beginPath();x.moveTo(212+a*4,46+b*16);x.quadraticCurveTo(212+a*4,46+b*4,212+a*16,46+b*4);x.stroke()})}
if(X.sh){x.fillStyle=X.sh===1?"#111":X.sh;x.fillRect(46,92,76,42);x.fillRect(134,92,76,42);x.fillRect(120,98,16,8);x.strokeStyle="rgba(255,255,255,.5)";x.lineWidth=4;x.beginPath();x.moveTo(58,102);x.lineTo(80,102);x.moveTo(146,102);x.lineTo(168,102);x.stroke()}
if(X.ms){x.fillStyle=INK;x.beginPath();x.moveTo(128,150);x.bezierCurveTo(112,138,84,142,70,162);x.bezierCurveTo(88,154,108,160,128,162);x.bezierCurveTo(148,160,168,154,186,162);x.bezierCurveTo(172,142,144,138,128,150);x.fill()}
if(X.ep){x.fillStyle=x.strokeStyle=INK;x.lineWidth=6;x.beginPath();x.moveTo(56,104);x.lineTo(2,88);x.moveTo(118,90);x.lineTo(254,56);x.stroke();x.beginPath();x.ellipse(88,112,33,28,-.3,0,7);x.fill()}
if(X.mn){x.strokeStyle="#d8a31a";x.lineWidth=5;x.beginPath();x.arc(168,112,36,0,7);x.stroke();x.beginPath();x.moveTo(198,132);x.quadraticCurveTo(214,180,202,236);x.stroke()}
if(X.hn){x.fillStyle="#d1202a";x.strokeStyle=INK;x.lineWidth=4;[[34,1],[222,-1]].forEach(([hx,s])=>{x.beginPath();x.moveTo(hx-s*6,52);x.lineTo(hx-s*2,6);x.lineTo(hx+s*30,36);x.closePath();x.fill();x.stroke()})}
if(X.ha){x.strokeStyle="#ffd23f";x.lineWidth=8;x.beginPath();x.ellipse(128,16,46,11,0,0,7);x.stroke()}
if(gl||X.gs){x.strokeStyle="#222";x.lineWidth=5;[88,168].forEach(bx=>{x.beginPath();x.arc(bx,112,35,0,7);x.stroke()});x.beginPath();x.moveTo(123,110);x.lineTo(133,110);x.stroke()}}

/* ---------- BUILD ---------- */
function patTex(p,c){const cv=document.createElement("canvas");cv.width=cv.height=128;const x=cv.getContext("2d");x.fillStyle=x.strokeStyle=c;x.lineWidth=8;x.lineCap=x.lineJoin="round";
switch(p){
case"band":x.fillRect(0,50,128,28);break;
case"stripes":for(let y=14;y<128;y+=32)x.fillRect(0,y,128,14);break;
case"star":star(x,64,66,50,22);x.fill();break;
case"heart":heart(x,64,62,38);x.fill();break;
case"logo":x.fillRect(30,36,68,56);x.fillStyle="#fff";x.beginPath();x.moveTo(56,52);x.lineTo(82,64);x.lineTo(56,76);x.fill();break;
case"pocket":x.lineWidth=6;x.strokeRect(70,30,40,38);break;
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case"badge":star(x,86,44,28,12);x.fill();x.strokeStyle="#9a6b00";x.lineWidth=4;x.stroke();break;
case"buttons":[26,62,98].forEach(y=>{x.beginPath();x.arc(64,y,8,0,7);x.fill()});break;
case"bolt":x.beginPath();x.moveTo(78,8);x.lineTo(40,68);x.lineTo(62,68);x.lineTo(50,120);x.lineTo(92,54);x.lineTo(68,54);x.closePath();x.fill();break;
case"tie":x.beginPath();x.moveTo(54,4);x.lineTo(74,4);x.lineTo(70,32);x.lineTo(84,98);x.lineTo(64,122);x.lineTo(44,98);x.lineTo(58,32);x.closePath();x.fill();break;
case"smile":x.beginPath();x.arc(64,64,42,0,7);x.fill();x.fillStyle=x.strokeStyle="#222";x.fillRect(48,46,9,16);x.fillRect(71,46,9,16);x.lineWidth=6;x.beginPath();x.arc(64,66,22,.25,2.9);x.stroke();break}
return new THREE.CanvasTexture(cv)}
function mkHat(t,c){const g=new THREE.Group(),m=P(c),A=(ge,x=0,y=0,z=0,sc)=>{const o=M(ge,m,x,y,z);if(sc)o.scale.set(...sc);g.add(o)};
const Cy=(a,b,h,s=32)=>new THREE.CylinderGeometry(a,b,h,s),Sp=(r,half)=>new THREE.SphereGeometry(r,32,16,0,Math.PI*2,0,half?Math.PI/2:Math.PI),Co=(r,h,s=24)=>new THREE.ConeGeometry(r,h,s);
switch(t){
case"cap":A(Sp(.68,1),0,1.08,0,[1,.62,1]);A(Cy(.69,.69,.1),0,1.1);A(Cy(.42,.42,.06,40),.05,1.1,.62,[1,1,.75]);A(Sp(.085),0,1.52);break;
case"beanie":A(Sp(.7,1),0,1,0,[1,.85,1]);A(Sp(.16),0,1.6);break;
case"top":A(Cy(.9,.9,.08),0,1.15);A(Cy(.52,.52,.95),0,1.6);break;
case"crown":A(Cy(.62,.62,.28),0,1.28);for(let k=0;k<5;k++){const a=k*Math.PI*2/5;A(Co(.14,.4,8),Math.sin(a)*.5,1.62,Math.cos(a)*.5)}break;
case"wizard":A(Cy(.95,.95,.05),0,1.2);A(Co(.65,1.6,32),0,2);break;
case"helmet":A(Sp(.74,1),0,1,0,[1,.9,1.02]);break;
case"band":A(Cy(.68,.68,.16),0,1);break;
case"cowboy":A(Cy(1.05,1.05,.06),0,1.2);A(Cy(.55,.6,.55),0,1.5);break;
case"chef":A(Cy(.55,.5,.5),0,1.4);A(Sp(.62),0,1.95,0,[1,.8,1]);break;
case"party":A(Co(.42,1.05),0,1.7);A(Sp(.1),0,2.25);break;
case"police":A(Cy(.72,.72,.22),0,1.3);A(Cy(.82,.78,.12),0,1.46);A(Cy(.45,.45,.05,40),0,1.22,.66,[1,1,.7]);{const b=M(Sp(.11),P("#ffd23f"),0,1.32,.73);b.scale.set(1,1,.4);g.add(b)}break;
case"ears":A(G(.28,1.1,.16,.08),-.3,1.75);A(G(.28,1.1,.16,.08),.3,1.75);break}
return g}
const scene=new THREE.Scene(),cam=new THREE.PerspectiveCamera(40,1,.1,100),R=new THREE.WebGLRenderer({antialias:true,alpha:true});
R.shadowMap.enabled=true;R.shadowMap.type=THREE.PCFSoftShadowMap;document.body.prepend(R.domElement);
scene.add(new THREE.HemisphereLight(0xffffff,0x8899bb,.85));
const dl=new THREE.DirectionalLight(0xffffff,.85);dl.position.set(5,12,8);dl.castShadow=true;dl.shadow.mapSize.set(1024,1024);Object.assign(dl.shadow.camera,{left:-8,right:8,top:10,bottom:-4,near:1,far:40});dl.shadow.camera.updateProjectionMatrix();scene.add(dl);
const fl=new THREE.Mesh(new THREE.PlaneGeometry(40,40),new THREE.ShadowMaterial({opacity:.28}));fl.rotation.x=-Math.PI/2;fl.receiveShadow=true;scene.add(fl);
const disc=new THREE.Mesh(new THREE.CircleGeometry(3.4,48),new THREE.MeshBasicMaterial({color:0xffffff,transparent:true,opacity:.3}));disc.rotation.x=-Math.PI/2;disc.position.y=.01;scene.add(disc);

const S={c:0,e:1,o:1,p:0,talk:0,spin:0},cur=new Array(16).fill(0);let ch=null;
function build(){
if(ch){scene.remove(ch.root);ch.root.traverse(o=>{if(o.material){[].concat(o.material).forEach(m=>{m.map&&m.map.dispose();m.dispose()});o.geometry.userData.own&&o.geometry.dispose()}})}
const [,skC,hrC,hs,sx,sy,hz]=CH[S.c],[,shC,paC,soC,pat,patC,hat,hatC,long,shorts,ex,exC]=OU[S.o];
const root=new THREE.Group(),body=new THREE.Group();root.add(body);root.scale.set(sx*sy,sy,sx*sy);
const skin=P(skC),shirt=P(shC),pants=P(paC),shoe=P(soC);
body.add(M(G(2,2,1,.2),shirt,0,3,0));
if(pat){const m=new THREE.Mesh(new THREE.PlaneGeometry(1.5,1.5),new THREE.MeshBasicMaterial({map:patTex(pat,patC),transparent:true}));m.position.set(0,3,.505);body.add(m)}
const limb=(x,y,parts)=>{const g=new THREE.Group();g.position.set(x,y,0);parts.forEach(([ge,m,py])=>g.add(M(ge,m,0,py,0)));body.add(g);return g};
const arm=(x)=>{const g=new THREE.Group(),geo=limbGeo(.98,1.96,.98,.2);g.position.set(x,3.85,0);g.add(M(geo,[shirt,long?shirt:skin],0,-.93,0));g.geo=geo;body.add(g);return g};
const leg=(x)=>{const g=new THREE.Group(),geo=limbGeo(.97,1.8,.98,.16),k=new THREE.Group(),f=new THREE.Group();g.position.set(x,2,0);k.position.y=-.9;k.add(M(geo,[pants,shorts?skin:pants]));f.add(M(G(1,.3,1.08,.12),shoe,0,-.94,0));k.add(f);g.add(k);body.add(g);g.geo=geo;g.foot=f;return g};
const la=arm(1.5),ra=arm(-1.5),ll=leg(.5),rl=leg(-.5);
if(ex==="cape")body.add(M(G(1.9,2.6,.12,.05),P(exC),0,2.8,-.58));
if(ex==="pack")body.add(M(G(1.3,1.5,.5,.15),P(exC),0,3.2,-.75));
if(ex==="scarf")body.add(M(G(2.1,.4,1.1,.15),P(exC),0,3.95,0));
const head=new THREE.Group();head.position.set(0,4,0);head.scale.setScalar(hz);body.add(head);
head.add(M(G(1.22,1.16,1.16,.34),skin,0,.62,0));
const cv=document.createElement("canvas");cv.width=cv.height=256;const tex=new THREE.CanvasTexture(cv);tex.anisotropy=4;
const fm=new THREE.Mesh(new THREE.PlaneGeometry(1,1),new THREE.MeshBasicMaterial({map:tex,transparent:true,depthWrite:false}));fm.position.set(0,.58,.585);head.add(fm);
const hm=P(hrC),top=()=>head.add(M(G(1.3,.4,1.26,.15),hm,0,1.08,0));
if(hs===1){top();head.add(M(G(1.3,.25,.2,.1),hm,0,.98,.56))}
if(hs===2){top();head.add(M(G(1.3,1.5,.35,.15),hm,0,.45,-.55));[-1,1].forEach(s=>head.add(M(G(.2,.9,1,.1),hm,s*.66,.55,0)))}
if(hs===3){top();[-1,1].forEach(s=>head.add(M(G(.3,.9,.3,.12),hm,s*.78,.3,-.1)))}
if(hs===4){top();head.add(M(new THREE.SphereGeometry(.3,16,12),hm,0,1.5,-.1))}
if(hs===5)head.add(M(G(.25,.5,1.1,.1),hm,0,1.35,0));
if(hs===6){[-1,1].forEach(s=>head.add(M(G(.24,.62,.95,.1),hm,s*.67,.48,-.05)));head.add(M(G(1.22,.5,.22,.1),hm,0,.38,-.58))}
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if(hs===7){top();[-1,1].forEach(s=>head.add(M(new THREE.SphereGeometry(.36,16,12),hm,s*.7,.42,-.05)));head.add(M(new THREE.SphereGeometry(.42,16,12),hm,0,.62,-.5));head.add(M(new THREE.SphereGeometry(.26,16,12),hm,0,1.45,-.1))}
if(hat)head.add(mkHat(hat,hatC));
if(CH[S.c][0]==="Grampa")[-1,1].forEach(s=>{const mu=M(G(.28,.09,.1,.04),hm,s*.14,.5,.62);mu.rotation.z=-s*.25;head.add(mu)});
fcs.style.setProperty("--sk",skC);scene.add(root);ch={root,body,head,la,ra,ll,rl,ctx:cv.getContext("2d"),tex,fk:"",gl:ex==="glasses"}}
function face(t){const E=EM[S.e],bl=t%3.6>3.48&&E[1][0]!=="u";let mo=null,q=0;
if(S.talk){q=Math.round((.12+.35*Math.abs(sn(t*9)))*6)/6;mo=["talk",q]}
const k=S.e+"|"+(bl?1:0)+"|"+q;if(k===ch.fk)return;ch.fk=k;drawFace(ch.ctx,E,bl,mo,ch.gl);ch.tex.needsUpdate=true}

/* ---------- UI ---------- */
const ui=$("#ui"),setters={};
const fcs=document.createElement("div");fcs.id="faces";ui.appendChild(fcs);
{const tc=document.createElement("canvas");tc.width=tc.height=256;const tx=tc.getContext("2d");
EM.forEach((E,i)=>{const b=document.createElement("button"),c=document.createElement("canvas");c.width=c.height=72;b.setAttribute("aria-label",E[0]);b.title=E[0];drawFace(tx,E,false,null,false);c.getContext("2d").drawImage(tc,0,0,72,72);b.appendChild(c);b.onclick=()=>setters.e(i);fcs.appendChild(b)})}
const mark=()=>{[...fcs.children].forEach((b,j)=>b.classList.toggle("on",j===S.e));const b=fcs.children[S.e];fcs.scrollTo({left:b.offsetLeft-fcs.offsetLeft-fcs.clientWidth/2+b.offsetWidth/2,behavior:"smooth"})};
[["Character",CH,"c"],["Emotion",EM,"e"],["Outfit",OU,"o"],["Pose",PO,"p"]].forEach(([lab,arr,k])=>{
const d=document.createElement("div");
d.innerHTML=`<label>${lab} <span>${arr.length}</span></label><div class="r"><button aria-label="Previous ${lab}">‹</button><select aria-label="${lab}">${arr.map((a,i)=>`<option value="${i}">${a[0]}</option>`).join("")}</select><button aria-label="Next ${lab}">›</button></div>`;
const [bp,bn]=d.querySelectorAll("button"),sl=d.querySelector("select");
const set=i=>{S[k]=(i+arr.length)%arr.length;sl.value=S[k];if(k==="c"&&CH[S.c][7]!=null&&S.o!==CH[S.c][7]){S.o=CH[S.c][7];ui.querySelectorAll("select")[2].value=S.o}if(k==="c"||k==="o")build();if(k==="e")mark()};
sl.onchange=()=>set(+sl.value);bp.onclick=()=>set(S[k]-1);bn.onclick=()=>set(S[k]+1);setters[k]=set;ui.appendChild(d)});
const bar=document.createElement("div");bar.id="bar";
bar.innerHTML=`<button id="rnd">Randomize</button><button id="tk">Talk</button><button id="sp">Spin</button>`;ui.appendChild(bar);
$("#rnd").onclick=()=>{setters.c(Math.random()*CH.length|0);setters.e(EM.length*Math.random()|0);setters.o(Math.random()*OU.length|0);setters.p(Math.random()*10|0)};
[["#tk","talk"],["#sp","spin"]].forEach(([id,key])=>{const b=$(id);b.onclick=()=>{S[key]^=1;b.classList.toggle("on",!!S[key])}});
window.rig={state:S,setCharacter:i=>setters.c(i),setEmotion:i=>setters.e(i),setOutfit:i=>setters.o(i),setPose:i=>setters.p(i)};

/* ---------- CAMERA + LOOP ---------- */
let th=.35,ph=.12,dist=19;const pts=new Map();let pd=0;const cvs=R.domElement;
cvs.onpointerdown=e=>{cvs.setPointerCapture(e.pointerId);pts.set(e.pointerId,[e.clientX,e.clientY])};
cvs.onpointermove=e=>{const o=pts.get(e.pointerId);if(!o)return;if(pts.size===1){th-=(e.clientX-o[0])*.01;ph=clamp(ph+(e.clientY-o[1])*.006,-.1,1.2)}
pts.set(e.pointerId,[e.clientX,e.clientY]);if(pts.size===2){const[a,b]=[...pts.values()],d=Math.hypot(a[0]-b[0],a[1]-b[1]);if(pd)dist=clamp(dist*pd/d,7,32);pd=d}};
cvs.onpointerup=cvs.onpointercancel=e=>{pts.delete(e.pointerId);pd=0};
cvs.onwheel=e=>{dist=clamp(dist*(1+e.deltaY*.001),7,32)};
function resize(){R.setPixelRatio(Math.min(devicePixelRatio,2));R.setSize(innerWidth,innerHeight);cam.aspect=innerWidth/innerHeight;cam.updateProjectionMatrix()}
addEventListener("resize",resize);resize();
setters.e(1);build();let last=0;
function loop(ms){requestAnimationFrame(loop);const t=ms/1000,dt=Math.min(.05,t-last);last=t;
const tg=PO[S.p][1](t),k=Math.min(1,dt*9);for(let i=0;i<16;i++)cur[i]+=((tg[i]||0)-cur[i])*k;
const c=ch;c.la.rotation.set(cur[0],0,cur[1]);c.ra.rotation.set(cur[2],0,cur[3]);c.ll.rotation.x=cur[4];c.rl.rotation.x=cur[5];bend(c.la.geo,Math.min(0,cur[12]));bend(c.ra.geo,Math.min(0,cur[13]));c.ll.foot.rotation.x=Math.max(0,cur[14]);bend(c.ll.geo,c.ll.foot.rotation.x);c.rl.foot.rotation.x=Math.max(0,cur[15]);bend(c.rl.geo,c.rl.foot.rotation.x);
c.head.rotation.set(cur[6],cur[7],cur[8]);c.body.position.y=cur[9];c.body.rotation.x=cur[10];c.body.rotation.y=cur[11];
if(S.spin)c.root.rotation.y+=dt*.9;face(t);
cam.position.set(dist*Math.sin(th)*Math.cos(ph),.6+dist*Math.sin(ph),dist*Math.cos(th)*Math.cos(ph));cam.lookAt(0,.6,0);R.render(scene,cam)}
requestAnimationFrame(loop);
</script></body></html>
```